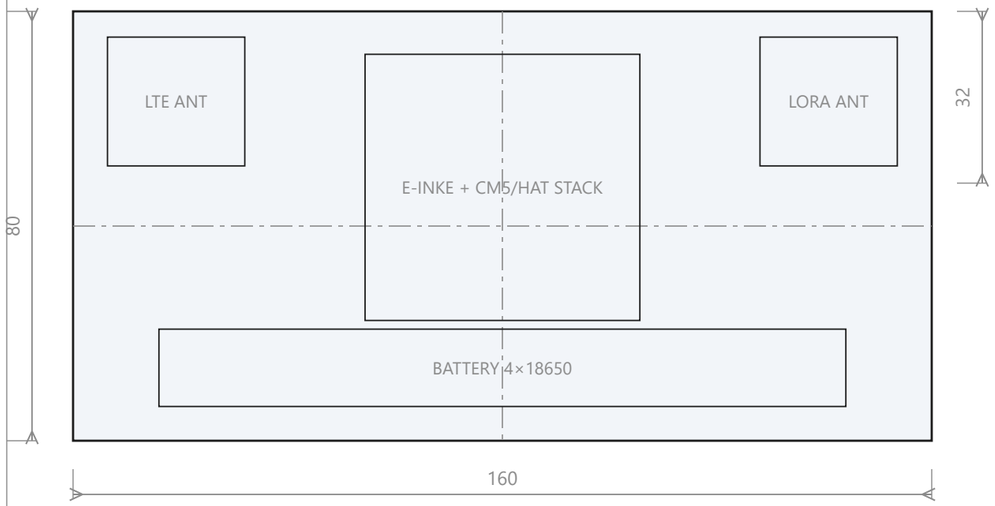
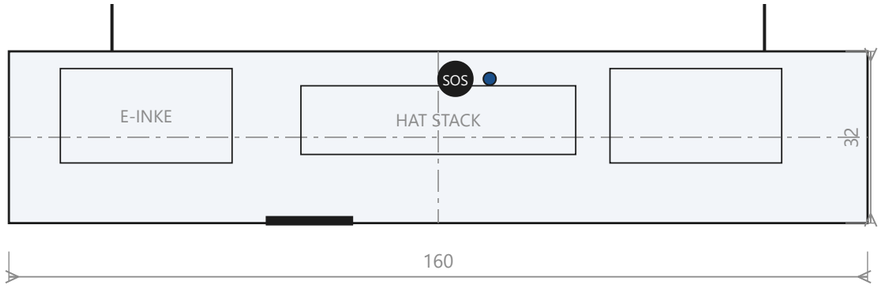
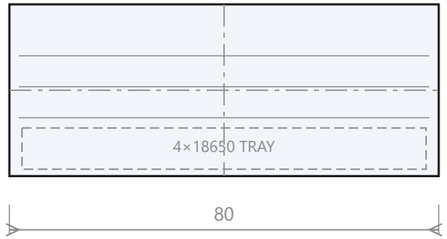
TOP VIEW



FRONT VIEW



SIDE VIEW



scale: shown NTS · dims in mm

WHAT THIS NODE IS

— Handheld brick = mailbox + mesh repeater + public bulletin (E-Ink) + SOS beacon.

— Runs ArcticChain agent on :8080 — SQLite store-and-forward, signed Ed25519 envelopes.

— LoRa 915 MHz mesh to neighbors; LTE backhaul only when a tower exists.

— Works with zero connectivity — that is the product.

KEY FEATURES

FEATURE	DETAIL
Display	7.5in e-Paper 800×480 — sunlight readable, zero idle power
Mesh radio	SX1262 LoRa 915 MHz on SPI1 (GPIO 16–21)
Backhaul	SIM7600 LTE HAT (optional) + GNSS + BT
Power	4×18650 → 14.8 V → 5 V USB-C PD · ~44 Wh
Enclosure	160×80×32 mm printed PETG · ~450 g assembled

ACCEPTANCE TEST

— Unit A POST /message → Unit B GET /inbox — with LTE in airplane mode.

— Mesh delivery with no towers = the whole pitch in one test.

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DRAWN **Surafel Negussie**
SHEET **1/5**
DATE **2026-11-12**
DWG NO **AF-L1-101**
SCALE **NTS**
UNITS **mm**

PARTS LIST — PER UNIT (ORDER ×2 FOR ACCEPTANCE TEST)

#	PART	SUPPLIER	APPROX CAD	QTY
1	Raspberry Pi CM5 4GB + CM5 IO Board	PiShop / Canada Computers	$90–130	1
2	7.5in e-Paper HAT 800×480 SPI	Waveshare / DFRobot	$80–110	1
3	SIM7600 4G/LTE HAT (GNSS+BT ok)	DFRobot / Waveshare	$60–90	1
4	SX1262 LoRa breakout 915 MHz (E22-900M30S class)	DFrobot / AliExpress	$15–25	1
5	Magnetic pogo-pin connector pairs ×6	AliExpress / Amazon.ca	$30–60	1
6	4×18650 flat-top protected + 14.8→5 V 3 A buck	Electronics supplier	$40–70	1
7	PETG filament (both shells + trays ~200 g)	Any filament supplier	$25–35	0.5

POWER TREE

4×18650 → 14.8 V · 44 Wh
14.8→5 V BUCK → 3 A · USB-C PD
5 V RAIL → fuse 5 A · 10 A in

Runtime est: days–weeks depending on LTE duty. Rev-B adds solar charge port and LiFePO4 cells for -45 °C service.

GPIO ALLOCATION

SIGNAL	GPIO	NOTES
E-Ink SPI	9/10/11/8	spi0 · HAT default
E-Ink busy/rst	25 / 17	poll before refresh
LoRa SPI1	16–21	dtoverlay=spi1-3cs — NOT spi0
LoRa rst/DIO1	24 / 23	interrupt RX
LTE modem	USB	CDC-ACM preferred
SOS button	27	pull-up input
Status LED	22	PWM

/BOOT/FIRMWARE/CONFIG.TXT — REQUIRED

```
dtparam=i2c_arm=on
dtparam=spi=on
dtparam=uart0=on
dtoverlay=spi1-3cs
dtoverlay=gpio-shutdown
disable_overscan=1
```

E-Ink = spi0 (GPIO 7–11) · LoRa = spi1 (GPIO 16–21). Never share a bus.

STACK ORDER (BOTTOM → TOP)

— USB-C power + battery tray (bottom bay)
— CM5 IO Board — power, HDMI, USB, GPIO
— 4G/LTE HAT — UART + USB AT commands
— E-Ink HAT — SPI0 + GPIO25 busy
— LoRa SX1262 — SPI1 GPIO16–21 (top)

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SHEET	**2/5**
DATE	**2026-11-12**
DWG NO	**AF-L1-102**
SCALE	**NTS**
UNITS	**mm**

ASSEMBLY SEQUENCE

1. Print chassis_shell.stl + battery_tray.stl — PETG, 40% infill, 4 walls.
2. Flash Pi OS Lite 64-bit to SD (Imager → SSH on, user 'arctic').
3. Bench-test bare CM5 first: boot, SSH, i2cdetect -y 1.
4. Stack HATs per sheet AF-L1-102; wire LoRa to GPIO 16–21 (spi1).
5. Wire power: 4×18650 → buck → USB-C; fuse 5 A rail, 10 A input.
6. Deploy agent: arcticchain_agent.py + systemd unit (poc/SETUP.md §5).
7. Acceptance test: POST /message on A → GET /inbox on B, LTE off.

AGENT API — :8080

ENDPOINT	DETAIL
GET /health	lora/lte status, uptime, queue depth
GET /inbox	pending messages (JSON)
GET /signal	mesh signal metrics
POST /message	{to, text} — queues or sends via LoRa

KNOWN GAPS (BY DESIGN)

— LoRa send/poll are TODOs — plaintext beacon until SX1262 driver lands (wk 3).
— E-Ink rendering is a stub; real Waveshare driver is Level-2.
— No power-path controller — battery swap reboots CM5 (acceptable PoC).
— Rev-B targets: -45 °C cells, IP65 gasket, glove SOS, solar input.

FIELD ACCEPTANCE CRITERIA

TEST	PASS CONDITION
Mesh exchange	message delivered A→B with LTE disabled, ≤60 s
Range	≥3 km line-of-sight at default SF
Cold soak	operates -20 °C (rev-B: -45 °C)
Power loss	SQLite inbox survives hard power cut

FILES IN REPO

— cad/level1_chassis.py → cad/level1/*.stl + dimensions.json (validated)
— poc/arcticchain_agent.py · arcticchain-agent.service · SETUP.md · WIRING.md
— docs/ORDER_LIST.md · BUILD_GUIDE.md · HARDWARE.md
— firmware-hal · mesh-net · compositor · simulator (Rust, 46 tests)

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DWG NO **AF-L1-103**
SCALE **NTS**
UNITS **mm**

SPI / BUS MAP

BUS	DEVICE	PINS	SPEED	NOTES
SPI0	E-Ink 7.5in	GPIO 9/10/11 · CE0=8	8 MHz	Waveshare HAT default
SPI0 ctrl	busy / reset	GPIO 25 / 17	—	poll BUSY before refresh
SPI1	SX1262 LoRa	GPIO 16–19 · CE1=17?	8 MHz	dtoverlay=spi1-3cs
SPI1 ctrl	NSS/RST/BUSY/DIO1	GPIO 20/21/22/23	—	DIO1 = RX interrupt
I2C	PMIC / sensors	GPIO 2/3 (i2c_arm)	400 kHz	i2cdetect -y 1
UART	LTE modem	USB CDC-ACM	115200	preferred — no GPIO cost
GPIO	SOS / LED	GPIO 27 / 22	—	pull-up input / PWM out

SX1262 RADIO CONFIG (915 MHZ ISM)

PARAM	VALUE	PARAM	VALUE
Frequency	915–928 MHz	Spreading factor	SF7–SF12 (adaptive)
Bandwidth	125 kHz	Coding rate	4/5 – 4/8
TX power	+22 dBm max	Preamble	8 symbols
CAD	enabled	CAD duration	2 symbols
Duty cycle	<1% target	CAD → RX window	~200 ms

Mesh envelope = signed mesh-net frame (~200 B max). Broadcast beacon every 30–60 s; unicast relay on demand. CRC + whitening on.

BRING-UP ORDER

— 1. Boot bare CM5 — SSH, i2cdetect, vcgencmd get_throttled.
— 2. spi0 E-Ink — Waveshare python lib, show test pattern.
— 3. spi1 SX1262 — send/recv test with second unit (or SDR).
— 4. LTE HAT — AT init, signal query, data session.
— 5. Agent — systemd up, /health green, acceptance test.

POWER SEQUENCING

1 · BAT
4×18650 → 14.8 V → 5 V buck 3 A

⩛

2 · 5V RAIL
fuse 5 A → CM5 USB-C PD

⩛

3 · HATS
LTE HAT 5 V · E-Ink 3V3 rail · LoRa 3V3

⩛

4 · PERIPH
LoRa EN · display reset · button pull-ups

All logic = 3V3. E-Ink needs 3V3+5V flash rails per Waveshare spec — HAT handles it. Never feed 5 V to SX1262.

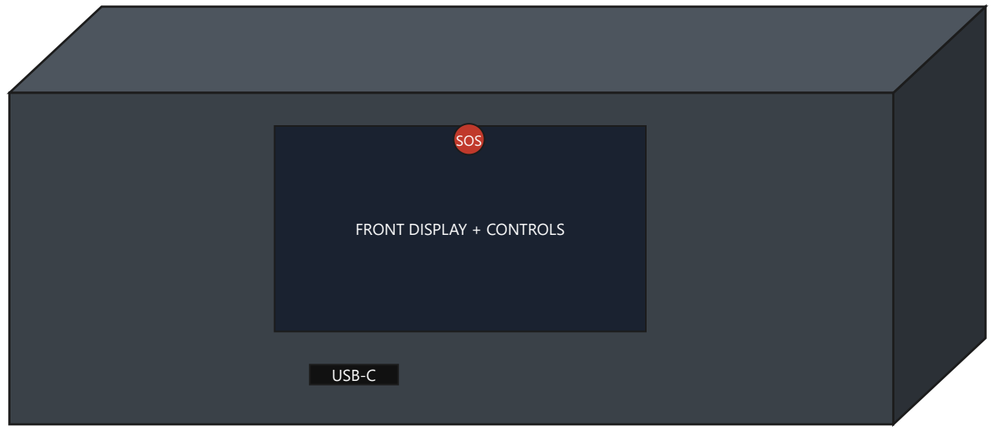
LEVEL / SAFETY NOTES

— SX1262 = 1V8–3V6 supply, 3V3 logic — do NOT 5 V tolerate.
— LoRa NSS on GPIO20 conflicts nothing on spi1-3cs — verify DT overlay at boot.
— E-Ink BUSY must be polled — refresh takes 3–20 s depending on waveform.
— USB-C PD negotiates 5 V/3 A; barrel alternative = 14.8→5 V buck only.
— Static-safe layout: keep 915 MHz trace <30 mm, 50 Ω controlled.

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SHEET	**4/4**
DATE	**2026-11-12**
DWG NO	**AF-L1-104**
SCALE	**NTS**
UNITS	**mm**



L1 BRICK — REV A

160 × 80 × 32 mm · ~450 g · matte textured

COLOR / FINISH OPTIONS

GRAPHITE
default — hides dirt

HI-VIS ORANGE
search & rescue

SAND
desert/arid sites

MATERIALS SPEC

COMPONENT	MATERIAL	FINISH	NOTE
Shell + tray	PETG (or ASA -20°C)	matte, 0.2 mm layers	hand-printable, repairable
Gasket/bezel	TPU 95A	soft-touch	display window seal
Fasteners	M3 stainless + threadlock	black oxide	hand-tool service
Labels	laser-etched PET or engraved	—	no stickers — wipes clean
Antenna	SMA whip + u.fl	rubber boot	field-replaceable

DESIGN NOTES

— Matte texture = glare-free in sun, hides scratches, better grip with gloves.
— Hi-vis orange variant mandatory for fire-season deployments.
— No painted logos — laser-etched 'ARCTICCHAIN · EMEBET INC.' on rear.
— Color does NOT change internals — same BOM, different filament only.
— Rev-B: soft-touch TPU overmold option for consumer feel.

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DATE	**2026-11-12**
DWG NO	**AF-L1-105**
SCALE	**NTS**
UNITS	**mm**